

LARGE EHV TRANSMISSION SUBSTATIONS

Design Basis, Design Basis Accident, and Military Action Extension

DBA-ES-TX-FC2: Large EHV Transmission Substation Facility Class

A Child Document Under

DBA-ES: Electric Sector Design Basis Framework (v0.9)

DBA-EN-SF: Energy Sector Framework (v0.8)

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For review by TransCo engineers, NERC reliability analysts, FERC staff, and policy officials responsible for electric sector critical infrastructure protection.

Section 1 — Purpose, Scope, and Inheritance

1.1 Purpose

This document establishes the Design Basis for large extra-high-voltage (EHV) transmission substations within the Electric Sector Design Basis Series. It is the second facility-class instrument in the series, following DBA-HY-FC1 (large hydroelectric storage facilities), and the first instrument to address TransCo's physical infrastructure directly.

The document applies the DB → DBA → DBA-MA three-layer analytical progression to EHV transmission substations serving metropolitan ring configurations: a baseline Design Basis (DB) that establishes postulated events from equipment failure, natural hazards, and operational conditions; a Design Basis Accident (DBA) layer that bounds the non-adversarial consequence envelope and establishes acceptance criteria; and a military action extension (DBA-MA) that applies the four characteristics of deliberate adversarial action — intentionality, persistence, precision, and denial — to each postulated event category.

Large EHV transmission substations are the single facility class for which the combination of long-lead-time equipment exposure and the intentionality and precision characteristics of military action produces the most severe extended-consequence scenario in the electric sector. The large power transformer (LPT) is the governing vulnerability: a custom-manufactured component with replacement lead times of eighteen to twenty-four months under normal procurement conditions, no field repair path following physical destruction, and no strategic reserve adequate to cover simultaneous loss at multiple critical facilities. The RT_HVLLC (Recovery Timeline for High-Value Long-Lead Components) metric, established as a series standard in DBA-ES §6.2, is the primary consequence-duration metric throughout this instrument.

1.2 Facility Class Definition

A facility within the scope of DBA-ES-TX-FC2 must satisfy all three of the following criteria.

Voltage class. The substation operates at extra-high voltage, defined for this series as 345 kilovolts or above. Substations at 345 kV, 500 kV, and 765 kV are within scope.

Metropolitan ring configuration. The substation serves as a supply node in a metropolitan ring transmission configuration — a topology in which multiple EHV substations form a ring around a major load center, collectively supplying the load pocket through a combination of radial feeds and looped circuits.

CIP-014 designation or long-lead-time equipment trigger. The facility carries a NERC CIP-014 Tier 1 or Tier 2 designation, or satisfies Trigger 3 of DBA-ES §1.5 by containing large power transformers at EHV voltage with an RT_HVLLC of twelve months or more.

1.3 Evidentiary Anchors

Metcalf Sniper Attack (April 16, 2013). Unidentified individuals discharged rifle fire into the cooling radiators of seventeen large power transformers at PG&E’s Metcalf transmission substation near San Jose, California. The attack lasted approximately nineteen minutes. The Federal Energy Regulatory Commission subsequently commissioned a classified analysis concluding that a coordinated attack on approximately nine critical transmission substations could cause blackouts affecting most of the continental United States. Metcalf established that LPT external cooling systems are a viable precision-attack vector requiring minimal adversary capability and producing an RT_HVLLC-class consequence.

UK National Grid EHV Substation Attack (2024). A coordinated physical attack on a National Grid extra-high-voltage substation in England caused significant disruption to the regional electricity supply. The attack demonstrated that EHV substations at the voltage class addressed by this instrument are active targets in adversary planning.

1.4 Inherited Methodology

All sector-level methodology is inherited from DBA-ES: Electric Sector Design Basis Framework (v0.9). The following table identifies inherited elements and any FC2-specific adaptations. The governing foundational instruments — FD-BL (placement), FD-BL-D1 (mode of discharge), FD-BL-D2 (indication integrity), and FD-EV (envelope) — are applied in this annex by reference.

Element	Source	FC2-Specific Adaptation
DB → DBA → DBA-MA three-layer progression	DBA-ES §3.1	Inherited without modification
Postulated event categories: PE, SE, CE, XC	DBA-ES §3.5	FC2-specific events instantiated in Section 3
Bounding event methodology	DBA-ES §4.1	Inherited; LPT is the governing long-lead-time equipment
Three-tier consequence taxonomy	DBA-ES §6.1	Both cascade and direct-dependency Tier 3 pathways apply
RT_HVLLC extended consequence regime	DBA-ES §6.2	RT_HVLLC is the primary FC2 consequence-duration metric
MA extension trigger criteria	DBA-ES §1.5	FC2 satisfies Triggers 1, 2, and 3 by definition
Four MA characteristics	DBA-ES §5.2	Applied to each FC2 postulated event category in Section 5
Dual Design Basis architecture	DBA-ES §3.4	Physical DB (this document) + AI Governance DB interaction points in §3.6
MA applicability four-filter framework	DBA-ES-MA-SF v0.2 §1.4; DBA-ES-MA-QB	Instantiated in §5.1 of this document

	v0.1 §5	
MA communications design basis (CB-1 through CB-5)	DBA-ES-MA-QB v0.1 §5	Instantiated in §5.5 of this document

1.5 Position in Hierarchy

Document Number: DBA-ES-TX-FC2-001

Series: Electric Sector Design Basis Series — Facility Class Instruments

Parent Document: DBA-ES: Electric Sector Design Basis Framework (v0.9)

Series Grandparent: DBA-EN-SF: Energy Sector Framework (v0.8)

Position in hierarchy: Facility-class instrument. All sector-level methodology is inherited from DBA-ES. This document applies that methodology to large EHV transmission substations and develops the RT_HVLLC consequence-duration analysis specific to this facility class.

Section 2 — Facility Profile: Large EHV Transmission Substation

2.1 Physical Configuration

A large EHV transmission substation in a metropolitan ring configuration is a substantial fixed installation: multiple transformer banks, an extensive EHV bus structure, numerous circuit breakers and disconnect switches, protective relay systems, and a control building housing the protection panels, SCADA equipment, and communications systems that govern the facility's operation.

The large power transformer is the critical element that defines the FC2 consequence profile. A 500 kV autotransformer bank is a custom-engineered component manufactured to the specific voltage ratio, MVA rating, and impedance characteristics of its application. No two installations are identical in all parameters. The full replacement timeline from order to energization is twelve to twenty-four months under normal conditions.

The transformer oil system is a secondary physical vulnerability. Oil loss from the cooling radiators — the Metcalf attack vector — removes the cooling function and causes thermal shutdown within hours. Radiators are located on the exterior of the transformer tank and are not hardened against ballistic or explosive attack in standard installations.

2.2 Grid Topology Role

An EHV substation in a metropolitan ring configuration performs three grid functions simultaneously: it steps down EHV transmission voltage to sub-transmission voltage; it serves as a switching node in the ring topology; and it provides local reactive power support that maintains voltage in the load pocket.

The metropolitan ring topology creates a shared-infrastructure vulnerability. Each substation in the ring depends on the other ring nodes to maintain load-pocket supply under contingency conditions. If two or more nodes are simultaneously de-energized in a coordinated attack, the ring topology's contingency coverage is exhausted, and the load pocket is deprived of supply.

2.3 Regulatory Context

NERC CIP-014 (Physical Security) is the primary mandatory regulatory instrument applicable to FC2 facilities. CIP-014 addresses the physical security posture of individual facilities. It does not establish a design basis, does not address the RT_HVLLC consequence duration, and does not address the coordinated attack on multiple ring nodes simultaneously. This document supplies the design basis layer that CIP-014 does not provide.

Section 3 — Design Basis: FC2 Postulated Event Categories

3.1 Sector-Level Inheritance

The four sector-level postulated event categories — PE (Power Supply and Grid Interface), SE (Structural and Physical Integrity), CE (Control and Communications), and XC (Cross-Sector Cascade) — are inherited from DBA-ES §3.5 and instantiated below with FC2-specific events.

3.2 Category PE — Power Supply and Grid Interface Events

Code	Event	Initiating Condition	Governing Concern
PE-1	N-1 transmission loss	Loss of a single transmission element connected to the substation. Protection systems clear the fault and isolate the failed element.	Substation must remain operational under N-1 conditions. Protection coordination must clear the fault without sequential tripping.
PE-2	N-1-1 transmission loss	Loss of two transmission elements simultaneously or within the recovery window of the first.	Metropolitan ring substations must demonstrate load continuity under N-1-1 through switching, load shedding, or load transfer.
PE-3	Complete loss of high-side bus	All transmission connections to the EHV bus were simultaneously lost, or the bus faulted.	Bounding PE event. Load pocket supply eliminated. Station service continuity determines operational capability during the event.
PE-SS	Station service loss	Loss of AC station service power to control panels, relay panels, SCADA, and communications.	Loss of station service removes monitoring and control at the moment most likely to require operator action.
PE-4	Voltage excursion — high	Sustained voltage above the ANSI C84.1 Range B upper limit from loss of load, combined with reactive surplus.	Transformer insulation stress. Requires automatic reactive power management or manual intervention.
PE-5	Voltage excursion — low	Sustained voltage below the range, a lower limit from heavy load and reactive deficit.	Voltage collapse risk in the metropolitan load pocket if not arrested.

3.3 Category SE — Structural and Physical Integrity Events

Code	Event	Initiating Condition	Governing Concern
SE-1	LPT failure — internal fault	Internal fault within the LPT winding, core, or bushing resulting in differential relay operation.	RT_HVLLC governs if the fault requires replacement. Design basis must distinguish repairable-fault case (days) from replacement case (18–24 months).
SE-2	LPT failure — external	Physical damage to LPT from fire, flooding,	Bounding SE event. SE-2 is the design basis accident for FC2. RT_HVLLC = 18–24 months for

	physical damage	projectiles, or blast. External damage is more likely to require full replacement.	a destroyed 500 kV autotransformer.
SE-3	EHV circuit breaker failure	Failure to open on command during fault clearing or failure to close during switching.	Breaker failure escalates a bounded fault into a wider outage. EHV CB lead times 6–12 months.
SE-4	Seismic event	Ground motion exceeding the design basis earthquake for the facility location.	Post-seismic fire at an oil-filled transformer under conditions of impaired site access is the compound governing concern.
SE-5	Extreme wind/tornado	Wind event exceeding design wind speed for transmission structures.	Direct tornado strike consequence similar to SE-2 with a natural initiating mechanism.
SE-6	Compound natural hazard	Sequential or simultaneous natural hazard events. Governing case: seismic + wildfire-smoke in Western US corridor.	Establishes the upper bound of the non-adversarial extended-consequence regime.

3.4 Category CE — Control and Communications Events

Code	Event	Initiating Condition	Governing Concern
CE-1	Loss of substation SCADA	Loss of communications between the substation and CntlCo EMS.	CntlCo loses situational awareness. Operational consequence: Substation continues passive operation under local protection.
CE-2	Protective relay misoperation	Relay operation not correctly corresponding to actual fault condition: false trip or failure to trip.	False trips initiate load shedding without a physical fault. Failure to trip allows the fault to persist until backup clearing.
CE-3	EMS failure at CntlCo	Loss or degradation of the CntlCo energy management system, including the state estimator.	Grid management capability degraded under concurrent physical stress at FC2 facilities.
CE-4	Cyber intrusion into substation controls	Unauthorized access to substation automation, relay settings, or SCADA RTUs.	Active intrusion modifying relay settings is analytically equivalent to CE-2 with adversarial intent.

3.5 Category XC — Cross-Sector Cascade Events

Code	Event	Initiating Condition	Governing Concern
XC-1	Metropolitan load pocket deprivation	FC2 loss removes the primary transmission supply to a metropolitan load pocket.	Cascade pathway to Tier 3 runs through DisCo substations. Water treatment, hospitals, and defense-critical loads are the consequence targets.
XC-2	Grid cascade from FC2 loss in the constrained	FC2 loss under pre-existing constraint triggers automatic protection actions	2003 Northeast Blackout paradigm. Design basis must establish conditions under which FC2 loss transitions to cascade initiator.

	corridor	propagating across the interconnection.	
XC-3	Defense-critical load disruption via direct supply	FC2 loss directly removes EHV supply to military installations or continuity-of-government sites.	Direct-dependency Tier 3 pathway. Requires service-area critical load inventory (DI-4).
XC-4	Gas-electric cascade from compressor auxiliary loss	FC2 loss reduces electric supply for compressor station auxiliaries, reducing gas pressure and curtailing gas-fired generation.	Self-reinforcing bidirectional cascade. FC2 design basis must identify gas-fired generation and pipeline assets in the service area.

3.6 Dual Design Basis Architecture

Consistent with DBA-ES §3.4, an FC2 facility simultaneously carries a sector-specific physical Design Basis and an AI Governance Design Basis. Three AI governance interaction points are established for FC2, closing DI-5.

EMS and state estimation — Bright Line: below. ML-augmented state estimation and contingency analysis tools are deterministic and formally testable; autonomous operation is permitted for state estimation and contingency screening. Bright Line condition: ML output must remain advisory; no automatic execution of dispatch commands without operator confirmation. State estimator input data loss (CE-1) triggers manual operations mode activation, not automatic ML substitution.

Protection relay setting optimization — Bright Line: below, with provenance requirement. ML relay setting optimization tools operate below the Bright Line. Provenance requirement: settings must be validated against an independent power system model before implementation; training data provenance must be documented and auditable; a model whose training data includes adversarially manipulated topology cannot be treated as below-Bright-Line until data integrity is confirmed.

Operational AI under degraded conditions — Bright Line: above; Command Broker required. Gen AI supporting switching sequence recommendations, contingency analysis, or emergency restoration guidance operates above the Bright Line. A qualified substation operator or CntlCo-designated switching authority is the mandatory Command Broker. Under MA denial conditions, the default state under communications loss is no-action, not autonomous Gen AI execution.

Section 4 — Design Basis Accidents: Non-Adversarial Envelope

4.1 Bounding Event Methodology

For FC2, the governing case within SE is unambiguous: SE-2 (LPT external physical damage requiring replacement) produces the highest consequence at every tier and does so on an RT_HVLLC-class duration. SE-2 is the design basis accident for FC2.

4.2 Non-Adversarial Consequence Envelope

Tier 1 — Facility-level. Loss of all EHV transformer capacity for the RT_HVLLC duration (18–24 months). Sub-transmission capacity may remain partially available if the low-voltage bus structure is undamaged. Station service is maintained through UPS for the UPS design duration.

Tier 2 — Grid-level. Metropolitan load pocket deprivation for a duration governed by alternate supply path availability through the ring topology. Complete load pocket deprivation occurs when FC2 loss eliminates all ring supply paths simultaneously.

Tier 3 — Cross-sector. Exhaustion of backup power posture at the most time-constrained dependent critical facility in the service area. If the most time-constrained load has a backup posture of less than 72 hours, the Tier 3 criterion must be satisfied through a combination of grid restoration and alternative supply arrangements.

4.3 Extended Consequence Regime

The non-adversarial DBA for FC2 must formally address the extended consequence regime defined in DBA-ES §6.2. An LPT replacement-class event produces a consequence persisting on the RT_HVLLC timeline regardless of whether the initiating event was adversarial. The analysis must identify which transmission constraints are binding during the outage, what compensating measures are available, and which cross-sector dependency thresholds are at risk within the restoration window.

Section 5 — Military Action Extension

5.1 MA Extension Applicability — Four-Filter Assessment

Per DBA-ES-MA-SF v0.2 §1.4 and DBA-ES-MA-QB v0.1 §5, each DBA-ES FC instrument must present a structured four-filter applicability assessment before the operator engages with the MA extensions. A conclusion of non-applicability requires documented reasoning against each filter and is a reviewable professional judgment. For FC2, the assessment is resolved at the class level as follows.

- **Filter 1 — Geolocation relative to defense-critical loads:** SATISFIED BY DEFINITION. EHV metropolitan ring substations supply all loads in their load pocket by definition, including defense-critical loads, military installations, and continuity-of-government facilities. XC-3 (defense-critical load disruption via direct supply) is an explicitly identified postulated event category for FC2.
- **Filter 2 — Position in strategic transmission corridors:** SATISFIED BY DEFINITION. FC2 facility class criteria require metropolitan ring configuration service. Metropolitan ring substations are by definition nodes in a strategic transmission corridor serving a major load center. Loss of an FC2 node in a constrained corridor is the XC-2 cascade initiator scenario.
- **Filter 3 — Substitutability:** LOW. The LPT's 18–24 month RT_HVLLC is the substitutability constraint. No spare transformer program exists that covers all 500 kV autotransformer configurations governing FC2 class facilities. STEP coverage gap at 500 kV custom configurations is registered as DI-2.
- **Filter 4 — Adversary targeting logic:** HIGH PRIORITY TARGET. DBA-ES-MA-SF v0.2 §4.2 identifies FC2 as a member of two of the three highest-consequence pattern attack combinations (FC2+FC3; FC2+GC-FC4). An adversary targeting maximum cascade effect will identify FC2 nodes as primary targets due to the RT_HVLLC persistence the LPT provides.

Conclusion: The MA design basis applies to all FC2 facilities. No facility-level exemption assessment is required. The MA extension sections of this instrument apply without a threshold qualification.

5.2 Application of the Four Characteristics

Category	Governing MA Scenario	Four Characteristics Applied	RT_HVLLC Implication	Acceptance Criterion
PE	Simultaneous multi-element transmission attack targeting the ring to isolate the load pocket	Intentionality: selects N-k combination for load pocket isolation. Precision: optimizes target selection. Denial: prevents switching operations and emergency	Not applicable to PE directly; RT_HVLLC governs if PE damages LPT.	Load pocket isolation must not produce unrecoverable cross-sector consequences within backup posture duration.

		restoration.		
SE	Precision kinetic attack on LPT at CIP-014 Tier 1 facility — Metcalf paradigm scaled to EHV	Intentionality: LPT selected as the highest-RT_HVLLC element. Persistence: 18–24 months lead time is effective persistence. Precision: cooling radiators and bushings targeted. Denial: security perimeter degradation prevents rapid damage assessment.	RT_HVLLC = 18–24 months for a destroyed 500 kV autotransformer. Governing consequence metric.	Physical hardening must reduce RT_HVLLC below the duration at which Tier 3 consequences become unrecoverable for the highest-dependency loads.
CE	Coordinated cyber-physical attack: relay setting modification combined with communications backhaul damage	Intentionality: relay settings modified for a specific failure mode. Persistence: modified settings persist until discovered. Precision: specific relay parameters targeted. Denial: SCADA loss prevents remote detection.	CE MA scenario removes operational infrastructure needed to manage the grid during RT_HVLLC-class SE event.	Relay settings must be verifiable through an independent out-of-band channel not subject to the same intrusion vector.
XC	Coordinated attack on FC2 and DisCo substations serving defense-critical loads in the same service area	Intentionality: targets both bulk supply and distribution last-mile simultaneously. Persistence: RT_HVLLC sustains isolation. Precision: target set maximizes defense-critical load overlap. Denial: prevents repair crews from reaching either tier.	RT_HVLLC at FC2 governs the upper bound; DisCo recovery governs the lower bound.	FC2 design basis must be coordinated with FC5 design basis for DisCo substations in the same service area.

5.3 The Metcalf Paradigm Scaled to EHV

The Metcalf attack demonstrated that precision physical attack on LPT cooling systems is viable at sub-EHV voltage. At EHV voltage, the consequence is qualitatively more severe for three reasons: EHV autotransformers have longer RT_HVLLC and smaller replacement manufacturing population; EHV metropolitan ring substations serve larger load pockets; and a coordinated ring attack includes both hardened CIP-014 EHV nodes and unsecured sub-transmission nodes whose vulnerability is not reduced by EHV-level hardening alone.

5.4 Coordinated Multi-Node Attack

The MA extension must address a coordinated attack on multiple nodes in the metropolitan ring simultaneously. A single-node attack can be partially absorbed through ring switching. A coordinated attack on two or more nodes exhausts contingency coverage and produces load pocket deprivation with no switching recovery path. This is the governing MA scenario — intentionality means the adversary will identify and target the minimum node set required to defeat the ring topology. For scenario-level modeling, see DBA-EN-UC-2 v0.1.

5.5 MA Communications Design Basis

DBA-ES-MA-QB v0.1 §5 establishes the MA communications design basis as a sector-level requirement applicable to all DBA-ES FC instruments. The MA communications baseline condition — primary SCADA and voice communications channels are unavailable — is the assumed design basis condition for all MA scenario analysis in this instrument. The following instantiates CB-1 through CB-5 for the FC2 facility class.

- **CB-1 (Backup channel):** FC2 facilities must maintain at least one backup communications channel to the serving CntlCo that is architecturally independent of the primary SCADA/EMS infrastructure. The channel must remain operational when primary channels are unavailable. Acceptable channels: dedicated licensed radio, satellite communications, or physically diverse fiber. Cellular backup does not satisfy CB-1 under MA conditions.
- **CB-2 (Local authority):** Substation operators must have pre-authorized authority to execute the FC2 emergency response procedures — including protective relay manual operation, transformer isolation, and station service load shedding — without CntlCo direction under conditions where the CntlCo is unreachable.
- **CB-3 (Command Broker continuity):** MA communications degradation does not suspend the Command Broker obligation for above-Bright-Line Gen AI advisory tools. The qualified substation operator or CntlCo-designated switching authority on duty remains the mandatory Command Broker. Above-Bright-Line Gen AI switching recommendations must not be actioned without licensed operator confirmation, regardless of communications status.
- **CB-4 (Verification cadence):** All backup communications channels must be tested at an interval specified in the facility’s design basis implementation. The test condition must simulate the MA baseline: primary SCADA and voice paths unavailable, backup channel as sole communications path.
- **CB-5 (Degraded-mode procedures):** FC2 emergency operating procedures must include a degraded-communications mode executable by substation operators without CntlCo direction. Procedures must address: (a) autonomous facility actions when CntlCo contact is lost; (b) backup channel activation sequence; (c) minimum safe operating state maintained pending communications restoration; (d) threshold for controlled de-energization rather than continued degraded-communications operation.

Section 6 — Consequence Analysis

6.1 Consequence Chain: SE-2 Governing Case

Initiating event. LPT cooling radiators penetrated by ballistic, explosive, or other physical attack. Transformer oil drains; Buchholz or sudden pressure relay initiates differential protection tripping.

Tier 1 — Facility. Transformer removed from service. The EHV bus serving the transformer's load is de-energized. Station service continuity depends on an alternate low-side supply path.

Tier 2 — Grid. Ring topology redistributes load to adjacent substations. If redistribution exceeds ring contingency capacity, load is shed in the load pocket.

Tier 3 — Cross-sector. Load shedding removes power from the DisCo distribution substations. Critical facilities transition to backup power. Tier 3 consequence escalates as backup posture is exhausted.

Extended consequence regime. LPT replacement initiated. Under normal procurement, RT_HVLLC is 18–24 months. The grid reliability posture during this period must be explicitly analyzed for each FC2 facility.

6.2 MA Extension Consequence Chain

The MA extension consequence chain differs from the non-adversarial chain in four ways. Intentionality extends the consequence from a single-node event to a multi-node event, targeting the minimum ring node set. Persistence extends the consequence duration through supply chain interdiction and access denial beyond the commercial RT_HVLLC baseline. Precision reduces adversary capability requirements by targeting the specific LPT element most likely to require replacement. Denial compounds all other factors by preventing damage assessment, repair crew access, and replacement equipment installation.

Section 7 — Acceptance Criteria

7.1 Tier 1 Acceptance Criteria

The facility must maintain station service continuity for a minimum of 72 hours following loss of all EHV and sub-transmission connections, through UPS or a dedicated independent station service supply. Protective relay settings must be verified to produce correct responses for all fault types and locations, verifiable through an out-of-band channel independent of the primary control system.

7.2 Tier 2 Acceptance Criteria

The metropolitan ring configuration must maintain load-pocket supply under N-1-1 contingency conditions. A single FC2 facility loss must not, under any pre-contingency operating condition within the TPL planning envelope, produce uncontrolled separation or cascading. The CntlCo must be capable of managing the grid in the FC2 facility's area during the first 72 hours following FC2 loss.

7.3 Tier 3 Acceptance Criteria

The maximum duration of power interruption to any defense-critical facility in the service area must not exceed that facility's backup power posture duration before mission-unrecoverable consequences occur. If the most time-constrained critical load has a backup posture of less than 72 hours, the Tier 3 criterion must be satisfied through a combination of grid restoration and alternative supply arrangements.

7.4 Extended Consequence Regime Acceptance Criteria

The grid reliability posture during the RT_HVLLC period must be analyzed and documented for each FC2 facility. The analysis must identify binding transmission constraints, available compensating measures, and cross-sector dependency thresholds at risk within the RT_HVLLC window. This is a site-specific design basis implementation requirement.

Section 8 — Deferred Items

Item	Description	Resolution Criterion
DI-1	Quantitative RT_HVLLC calibration. This document establishes RT_HVLLC at 18–24 months for a 500 kV autotransformer based on industry-reported lead times.	Updated RT_HVLLC range supported by manufacturer lead time data and STEP inventory analysis incorporated at next revision.
DI-2	STEP program coverage gap analysis. A formal gap analysis comparing FC2 transformer specifications against STEP inventory is needed to bound the population of FC2 facilities without a STEP-compatible spare.	STEP gap analysis completed; FC2 facilities without STEP coverage identified and logged as highest-priority hardening recommendation.
DI-3	Space weather / GIC postulated event. Geomagnetically induced currents from severe space weather are not yet addressed in this document.	GIC / space weather postulated event category added to SE events at next revision, with reference to NERC GMD standards.
DI-4	Service area defense-critical load inventory. XC-3 requires identification of defense-critical loads and their backup power posture. Cannot be completed at the series level.	Completed by each FC2 facility operator as part of site-specific design basis implementation.
DI-5	Dual-DB AI Governance interaction points. CLOSED v0.2. Bright Line positions and Command Broker requirements are established in §3.6.	CLOSED. Resolved in §3.6, v0.2.
DI-6	MA applicability four-filter block and communications design basis. CLOSED v0.3. Four-filter structured assessment established in §5.1; MA communications design basis CB-1 through CB-5 instantiated in §5.5.	CLOSED. Resolved in §5.1 and §5.5, v0.3. Addresses N-15 and N-16 batch pass requirements per DBA-ES-MA-SF v0.2 OI-8 and OI-1.

Section 9 — References

Regulatory and Standards References

Federal Energy Regulatory Commission. Order No. 693: Mandatory Reliability Standards for the Bulk-Power System. Washington, D.C.: FERC, 2007.

North American Electric Reliability Corporation. CIP-014-3: Physical Security. Atlanta: NERC. Current version.

North American Electric Reliability Corporation. TPL-001-5: Transmission System Planning Performance Requirements. Atlanta: NERC. Current version.

North American Electric Reliability Corporation. CIP-013-2: Supply Chain Risk Management. Atlanta: NERC. Current version.

North American Electric Reliability Corporation. FAC-001-3: Facility Connection Requirements. Atlanta: NERC. Current version.

Government and Policy References

Defense Science Board Task Force on DoD Energy Strategy. “More Fight — Less Fuel.” Washington, D.C.: Office of the Under Secretary of Defense, February 2008.

Department of Energy and North American Electric Reliability Corporation. Spare Transformer Equipment Program (STEP). Washington, D.C.: DOE. Current version.

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Wellinghoff, Jon, and Marcus Rutter. “Metcalf and the Vulnerability of Our Electric Grid.” Wall Street Journal, February 5, 2014.

Parfomak, Paul W. Physical Security of the U.S. Power Grid: High-Voltage Transformer Substations. Congressional Research Service Report R43604. Washington, D.C.: CRS, June 2014.

Series Parent and Cross-Reference Documents

Roxey, Tim. DBA-EN-SF: Energy Sector Framework. Version 0.8 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. DBA-ES: Electric Sector Design Basis Framework. Version 0.9 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. DBA-ES-MA-SF: Electric Sector Military Action Sector Framework. Version 0.2 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. DBA-ES-MA-QB: Electric Sector Military Action Quantitative Design Basis. Version 0.1 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. DBA-HY-FC1: Large Hydroelectric Storage Facilities. Version 0.4 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. DBA-EN-UC-2: Coordinated Multi-Node Attack on an Interconnection. Version 0.1 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. RT_HVLLC Strategic Reserve White Paper. Version 0.2 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. GenMix Optimization Whitepaper. Version 0.3 — DRAFT. Eclectic Technologies, 2026.

Revision History

Version	Date	Author	Description
0.1	June 2026	T. Roxey	Initial working draft. First publication under DBA-ES sector parent (v0.5). Established facility class definition (345–500 kV metropolitan ring configuration; CIP-014 Tier 1 or Tier 2; LPT with RT_HVLLC ≥ 12 months). Instantiated all four sector-level postulated event categories. Developed DBA non-adversarial consequence envelope. Applied DB $\rightarrow$ DBA $\rightarrow$ DBA-MA three-layer progression. Established RT_HVLLC as a governing consequence-duration metric. Incorporated Metcalf (2013) and UK National Grid EHV substation attack (2024) as primary evidentiary anchors.
0.2	June 2026	T. Roxey	Full three-layer instrument. Dual-DB AI Governance interaction points established in §3.6: EMS state estimation ML (below Bright Line, advisory only), protection relay ML optimization (below Bright Line with provenance requirement), Gen AI operational support (above Bright Line, Command Broker required). DI-5 closed.
0.3	June 2026	T. Roxey	N-15/N-16 batch pass. Three changes: (1) Parent updated from DBA-ES v0.5 to v0.9; series grandparent DBA-EN-SF v0.8 added to cover and §1.5. (2) §5.1 MA Applicability four-filter structured assessment added — all four filters satisfied at the class level; all-facilities conclusion stated. Closes N-15 / OI-8 for FC2. (3) §5.5 MA Communications Design Basis added — CB-1 through CB-5 instantiated for FC2 facility class. Closes N-16 / OI-1 for FC2. DI-6 added and marked CLOSED. DBA-EN-SF v0.8, DBA-ES-MA-SF v0.2, DBA-ES-MA-QB v0.1 added to §9 references. Revision history moved to the end of the document.